

TrES-1b

R-2275

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_110601 · created 2026-08-16T23:26:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455713.88648 +/- 0.00074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.181 +/- 0.046
Transit depth (Rp/Rs)^2	3.29 +/- 1.67 [%]
Orbital Inclination (inc)	85.17 +/- 1.54
Airmass coefficient 1 (a1)	1.208 +/- 0.043
Airmass coefficient 2 (a2)	-0.16 +/- 0.1
Scatter in the residuals of the lightcurve fit is	3.62 %
Best Comparison Star	#2 - [311, 215]
Optimal Aperture	2.86
Optimal Annulus	10.77
Transit Duration (day)	0.063 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.181 ± 0.046 vs 0.1358 (deviation 0.75σ)
Duration fit vs published	0.063 d vs 0.1042 d (deviation 0.95σ)
Mid-time O–C	1.0 min
Depth SNR	3.0
Residual scatter	3.62 %

Light curve

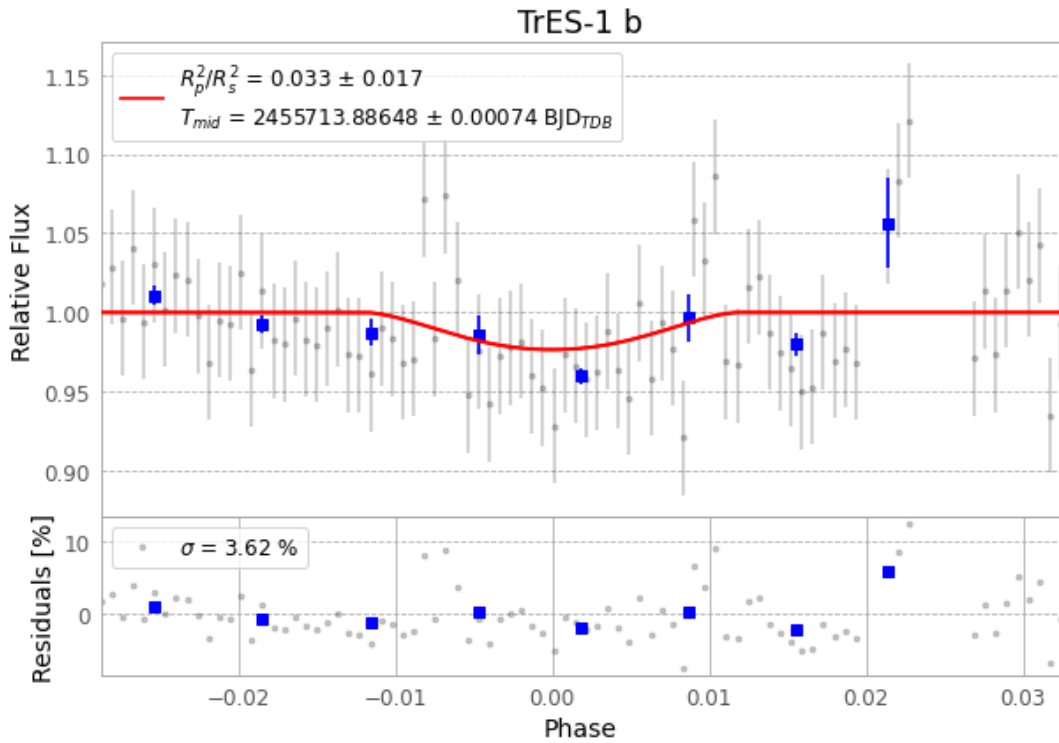


Figure 1 — FinalLightCurve_TrES-1 b_2011-06-01.png (SHA-256 4d7456670e52a072...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [240, 207], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9495dd5391bd0fb0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

90 FITS frames (59 MB), TRES-1110601070613.FITS → TRES-1110601113313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-110601.log (bea7f938cd6b...), FinalLightCurve_TrES-1 b_2011-06-01.png (4d7456670e52...), FinalLightCurve_TrES-1 b_2011-06-01.csv (bb24a834d090...)

Compute resources

Wall time 148.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 23:26Z.